



Diatoms abundant in ice-covered Lake Erie: An investigation of offshore winter limnology in Lake Erie over the period 2007 to 2010[☆]

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ABSTRACT

The limnology of offshore Lake Erie during periods of extensive (>70%) ice cover was examined from ship borne sampling efforts in 2007 to 2010, inclusive. Dense and discrete accumulations of the centric filamentous diatom *Aulacoseira islandica* (>10 µg Chl-*a*/L) were located in the isothermal (<1 °C) water column directly below the ice and only detectable in the ship wake; viable phytoplankton were also observed within ice. Evidence from these surveys supports the notions that winter blooms of diatoms occur annually prior to the onset of ice cover and that the phytoplankton from these blooms are maintained in the surface waters of Lake Erie and reduce silicate concentrations in the lake prior to spring. The mechanisms by which high phytoplankton biomass rise at this time of year requires further investigation, but these winter blooms probably have consequences for summer hypoxia and how the lake responds to climate change.

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Introduction

The biological limnology of the Laurentian Great Lakes and temperate lakes in general is known mostly from observations collected in late spring to late summer (cf. [Wetzel, 2001](#)). Whereas studies of ice extent, forecasting and the engineering aspects of ice in the Great Lakes have received considerable attention ([Assel et al., 2003](#); [Bolsenga, 1992](#)), research into winter limnology has been episodic, mainly attributed to the logistical difficulties associated with collecting samples during the winter. One issue that arises from this uneven data collection is that mechanistic models of lake ecosystem function must make assumptions of state and rate variables representative of the

lake at this time, or omit this time frame from the models. Consequently, our predictive capabilities are likely to be biased.

The most comprehensive study of winter limnology in the Laurentian Great Lakes to date is that by D.C. Chandler who coordinated a 5-year (1938–1942) study during which he observed “pulses” (blooms) of diatoms in mid-winter under the ice in the west basin of Lake Erie, low zooplankton biomass, and a variable light environment ([Chandler, 1940, 1942a, 1942b, 1944](#); [Chandler and Weeks, 1945](#)). Since the seminal work by Chandler, several monitoring efforts have contributed important insights into winter conditions in Lake Erie. They show a high degree of variability in biological and chemical properties but they require the consistency and scope of sampling effort to link this variability to ice cover, a dominant and consistent feature in Lake Erie in winter. [Glooschenko et al. \(1974\)](#) compared rates of winter primary production at open-water sites in Lakes Ontario and Erie, [Stewart \(1973\)](#) sampled all basins from a helicopter in 1972, and [Burns et al. \(1978\)](#) summarize limited sampling conducted by hovercraft in the east basin of Lake Erie in 1969, sampling of the central and east basins from an ice breaker in February, 1971 and a helicopter survey of the same basins in 1977. During the mid-1980s, scientists from the U.S. Environmental Protection Agency (EPA) Great Lakes National Program Office (GLNPO) and Argonne National Laboratory included a modest winter component to

[☆] We dedicate this manuscript to D.C. Chandler, the first president and founding member of the International Association for Great Lakes Research, for his seminal work in the 1930s on winter limnology of the Great Lakes.

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their seasonal water quality monitoring efforts on Lakes Michigan, Huron and Erie. This sampling was facilitated by use of a helicopter (Rockwell et al., 1989) from which they sampled 9 stations among the three basins of Lake Erie in January and February of 1985. Likewise, the Great Lakes Water Intake Monitoring program, an initiative of the Ontario Ministry of the Environment (OME) that began in 1976, supports weekly water intake biomonitoring from provincial sites that draw water from the Great Lakes and offers insights into temporal nutrient trends and phytoplankton biomass for nearshore locations including Lake Erie (Nicholls and Hopkins, 1993; Nicholls et al., 1980, 2001).

In response to the need for more hibernal limnology on the Great Lakes, members of the *ad hoc* research network MELEE (Microbial Ecology of Lake Erie Ecosystems) conducted a one week expedition in early November, 2001 on the east basin of Lake Erie. Phytoplankton were not nutrient (P) limited during this time but biomass was low (Wilhelm et al., 2003), suggesting that low light and temperature may have limited phytoplankton growth. Given the paucity of recent winter time investigations on this lake and the persistent summer-time phenomena of cyanobacterial blooms and the hypolimnetic hypoxia in the central basin of the lake, it raises the question that our comprehension of lake function in summer is limited by our awareness of plankton and nutrient dynamics throughout the entire year. Thus, the objective of the current study was to investigate Lake Erie during winter over a similar time period as Chandler's observations (4 years) but over a larger geographic range using modern instrumentation in tandem with classical limnological approaches to ascertain how the biological and chemical properties of the lake vary with ice cover. A working hypothesis was that algal production in winter may help fuel summer hypoxia and this investigation served to gather the information needed to support such a statement.

Materials and Methods

Sampling Locations

Hydrographic stations from each of the three basins in Lake Erie (Fig. 1) were sampled on cruises of the icebreaker CCGS *Griffon* during February 2007, February 2008, January and February 2009, and February 2010. Using a coast guard vessel for cruises of opportunity made accessing the same stations each year not possible due to the variable missions of the ship. Nevertheless, all basins in the lake were surveyed each year. Upon reaching a station, the ship would carefully open a small lead in the ice through which sampling could take place. All stations were sampled during daylight hours.

Surface Ice Cover Surveys

In February 2008, whilst underway, visible accumulations of algae under ice were enumerated in the 13 m wide wake of the ship at 12 knots by manual observation from the stern during daylight hours along a track recorded using navigational software (Offshore Navigator, ver. 5.07; Amesbury, MA). Times selected for doing this activity were based on availability of personnel. Thirteen transects (9 in central basin, 4 in west basin) were surveyed for a total of 1.33 km². Since some algae and seston were noticed in ice, freshly broken ice was collected from the surface of the lake by means of pikes and a polypropylene basket; ice was kept in a cooler for no longer than two hours before analysis.

Water Sampling

Water samples were collected using a 10-L Niskin bottle from two depths at each station (surface – 1 m below surface, and bottom – 2 m above sediment, unless otherwise stated) on a metered winch and by use of a submersible pump deployed 1 m below the surface. Water was processed onboard the ship within 1 hour of sampling for chemical and biological analyses (see below).

Untreated water samples from municipal water intakes in the west, central, and east basins (Fig. 1) over the study period were collected by municipal water treatment plant operators. The water intake pipes at the Union, Port Stanley, and Dunnville water treatment plants were 460 m, 1250 m and 488 m long, respectively; approximate depth of the water column above the pipes was 3, 8.5 and 7.5 m, respectively.

Zooplankton were collected from vertical tows using a 0.2 m diameter 153- μ m pore-size mesh Wisconsin-type plankton net. Tows were 10 m in depth with organisms collected on the down cast and upward tow.

Physical Analyses

A secchi disk (30 cm, white) was used to measure water clarity at each station, when possible. In 2008 and 2010, spectroradiometers were used to measure light extinction coefficients in water not covered directly by ice. Seven wavelengths between 379 nm and 665 nm were measured by Satlantic OCI200 (Halifax, NS) and UV wavebands 305, 320 and 340 nm, plus photosynthetically active radiation (PAR), were measured with a Biospherical BIC (San Diego, CA). Where the spectroradiometers were not available or could not be used, a Li-COR (Lincoln, NE) cosine quantum sensor and a metered

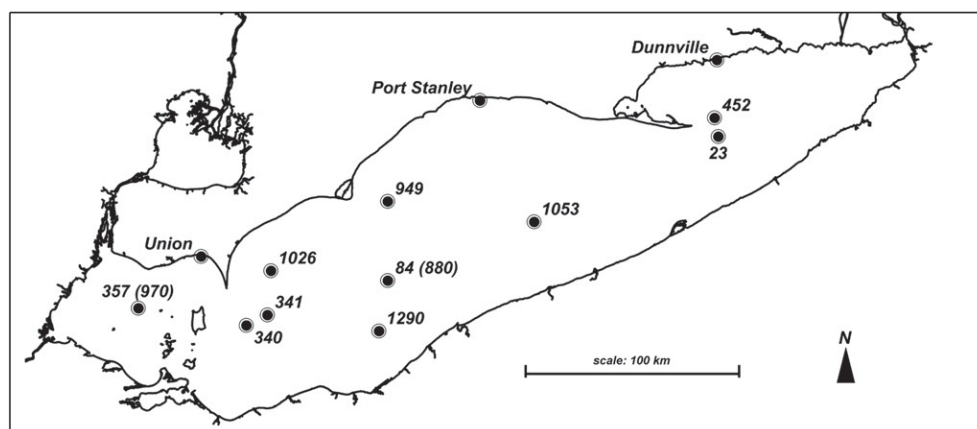


Fig. 1. Fixed hydrographic stations on Lake Erie sampled during winter investigative surveys in 2007, February 2008, 2009, and 2010. Public water intakes from which water quality data are available are shown as Union, Port Stanley (Elgin) and Dunnville located in the Province of Ontario. Parentheses refer to stations that are close to adjacent master stations.

line were used to measure PAR and derive extinction coefficients. Water column profiles of temperature, dissolved oxygen (DO), pH, conductivity and chlorophyll fluorescence were recorded during the 2008 and 2009 cruises using a Model 660 Sonde (YSI; Yellow Springs, OH) lowered at approximately 0.1 m/s. The sonde was not available for the 2007 cruise and failed during the 2010 cruise.

Chemical Analyses

Nutrients and seston elemental ratios were measured by the National Laboratory for Environmental Testing (NLET; Environment Canada, Water Science & Technology Directorate, Burlington, ON) using standardized techniques (NLET, 1994). Dissolved organic carbon (DOC) was analyzed using an Apollo 9000 Total Organic Carbon (TOC) analyzer (Teledyne Tekmar; Mason, OH) at 800 °C on water filtered through pre-rinsed 0.4- μ m nominal pore-sized glass fiber filters (Macherey-Nagel, GF-5). In 2007, particulate organic carbon (POC) and nitrogen (PON) was determined on seston collected on pre-combusted Whatman GF/F filters (nominally 0.7 μ m) using facilities at the University of Minnesota with a Perkin Elmer Series II CHN analyzer (Waltham, MA) using acetanilide as a standard. In 2008–2010, POC and PON were determined on the same model of CHN analyzer by NLET (NLET, 1994) on pre-combusted Whatman GF/C (nominally 1.2 μ m). Particulate phosphorus (PP) was measured colorimetrically following persulfate digestion as described previously (Hood et al., 2006). Chromophoric dissolved organic matter (CDOM) in water samples was measured by fluorometry (Twiss, 2011). Water samples collected at municipal water treatment plants were analyzed at the Ontario Ministry of the Environment laboratory for soluble reactive silicates reported as Si using standard analytical methods (Chow et al., 2010); detection limits were 0.02 mg/L for Si.

Biological Analyses

Plankton biomass and taxonomic composition were analyzed from samples collected from the Niskin bottle from surface waters in the lake. Raw water was transferred to bottles (amber) and fixed for later analysis (1% glutaraldehyde) or observed unfixed by microscopy on the ship. Taxonomic composition of fixed samples was determined by a combination of light, epifluorescence, and scanning electron microscopy (Booth, 1993; Carrick and Schelske, 1997). Picoplankton abundance measurements were made on samples (2 mL) concentrated onto 0.2- μ m nominal pore-size black polycarbonate filters (25-mm diameter; Poretics) and either stained with Acridine Orange for heterotrophic picoplankton (Hobbie et al., 1977) or unstained for phototrophic picoplankton (Fahnenstiel and Carrick, 1992). Nanoplankton were enumerated from slides prepared by filtering 10–20 mL onto darkened filters (0.8- μ m Nuclepore) subsequently stained with primulin (Caron, 1983). All slides were enumerated by counting >200 cells using a Leica DMR (Wetzlar, Germany) research microscope (1000 $\times$) equipped for autofluorescence (450–490 nm excitation and >515 nm emission), primulin analysis (320–380 nm excitation and >420 nm emission), and determination of phycoerythrin (green light 530–560 nm excitation and >580 nm emission). Dominant pigment fluorescence of individual cells was used to assign general taxonomic (phylum) position. Counting error for heterotrophic nanoplankton was estimated from Poisson counting statistics (Lund et al., 1958; Sandgren and Robinson, 1984).

Micro-plankton abundance and composition were determined from permanent mounts (Dozier and Richerson, 1975) by enumerating >400 cells. Aliquots (10–20 mL) were concentrated onto filters (3- μ m, Millipore), and filters were subsequently cleared with 50% glutaraldehyde while heating. Filters were then mounted on oversized slides using Permount. A transect on each slide was systematically scanned using phase contrast microscopy (400 $\times$). Coefficients of variation among counts of duplicate slides were typically <10%. In February

2008, duplicate samples were collected at 1 m in the water column at the mid lake station 1026. Samples were counted to estimate phytoplankton abundance and biomass according to the procedures described below. The sampling error for plankton abundance estimates were typical (coefficient of variation, 20.4%), and similar to those determined for samples collected during the spring to fall period in Lakes Erie (Carrick et al., 2005), Huron (Carrick and Fahnenstiel, 1989; Fahnenstiel and Scavia, 1987).

Algal taxonomy reported here conforms to that outlined by Prescott (1973); the phylogeny proposed by Rippka et al. (1979) was used for some of the cyanobacteria. Diatoms were identified using taxonomic keys of Patrick and Reimer (1975), as well as Krammer and Lang-Bertalot (1991a,b,c, 1997a,b). To confirm the diatom identification, samples obtained from net tows were cleaned in nitric acid to remove organic materials and air dried onto cover slips for light microscope and scanning electron microscope (SEM) analysis. Strewed diatom slides were made using Naphrax mounting medium and the diatoms analyzed using an Olympus BX51 Photomicroscope (Center Valley, PA) with high resolution Nomarski DIC optics. Samples, mounted on aluminum stubs and sputter coated with 10 nm of AuPd, were examined under a high resolution Hitachi S2700 SEM (Hitachi High Technologies America, Inc.; Schaumburg, IL).

Size-fractionated chlorophyll *a* (Chl *a*) concentrations were measured in duplicate for pico- (0.2–2 μ m), nano- (2–20 μ m), and micro-phytoplankton (>20 μ m) size classes (parallel filtrations), using the corresponding nominal pore-size, 47-mm diameter polycarbonate filters (GE Osmonics) after extraction of Chl *a* in 90% acetone (24 h at 4° C) and determined fluorometrically (Welschmeyer, 1994). In this study, Chl-*a* is used as a proxy for phytoplankton biomass in cases where we have not specifically estimated biomass by microscopy.

Physiological status of phytoplankton in lake water and from melted ice was assessed for phosphorus status (alkaline phosphatase activity; APA) and photosynthetic efficiency (variable Chl-*a* fluorescence). APA (E.C. 3.1.3.1.) was measured in unfiltered water samples collected by Niskin bottle from a depth of 1 m using a fluorimetric method (4-methyl umbelliferyl phosphate; Sigma; St. Louis MO) as described elsewhere (Sterner et al., 2004). Variable fluorescence of Chl-*a* in phytoplankton and suspensions of algae from melted ice was measured by Pulse Amplitude Modulated (PAM) fluorometry on dark adapted samples using a WATER-PAM Chl-*a* fluorometer (Heinz Walz GmbH; Effeltrich, Germany), with care to ensure that the instrument was properly blanked and that the fluorescence trace showed clear saturation and thus a good estimate of F_m . The dark-adapted variable fluorescence ratio of Chl-*a* (F_v/F_m) is a measure of the maximum quantum yield of photosystem II and an index of photosynthetic integrity. A PHYTO-PAM (Heinz Walz GmbH) with fiber optic light guides was additionally used to measure F_v/F_m of algae within sampled ice. Temperatures were maintained within 2–4 °C of ambient during measurements.

Bacterial abundance was determined in samples collected using Niskin bottles and immediately preserved in glutaraldehyde (final concentration, 2.5%). Samples (2 mL) for bacterial abundance measurements were collected onto 0.2- μ m pore-size black polycarbonate filters (25-mm diameter; Poretics) and stained with Acridine Orange (Hobbie et al., 1977). Bacterial cell counts were generated for true duplicates, except where otherwise noted, and presented as average $\pm$ the range.

Taxonomic composition and biomass of zooplankton was determined on organisms collected from a net and rinsed into a polyethylene cup and sugar formalin added to give a final concentration of 5% (v:v) formaldehyde and 2.5% (w:v) sucrose. Samples were refrigerated and kept dark. Subsamples were enumerated using the key of Balcer et al. (1984). In general, 50 to 100 individuals per sample were identified and enumerated at 50 $\times$ magnification using a Leica L2, research grade dissection microscope. The plankton net also collected filamentous

diatoms and was used at times to collect high densities for experimentation. Counting error for crustacean zooplankton was not estimated here; however, estimates were based on enumerating 1–2% of total water column net hauls.

Results

Ice Cover

Ice coverage on Lake Erie each February during the 2007–2010 ranged from 70 to 90% total lake cover (Fig. 2; data provided by G.A. Leshkevitch of the National Oceanic and Atmospheric Administration–Great Lakes Environmental Research Laboratory [NOAA–GLERL]). The most remarkable features observed on the ice covered lake during all cruises were the patches of discolored ice and water that proved (see below) to reflect the presence of silt and high concentrations of diatom algae (Fig. 3). We termed these discontinuous patches ‘caches,’ in recognition of their concealed nature and concentrated biomass, visibly obvious only if the ice is disturbed by the ship

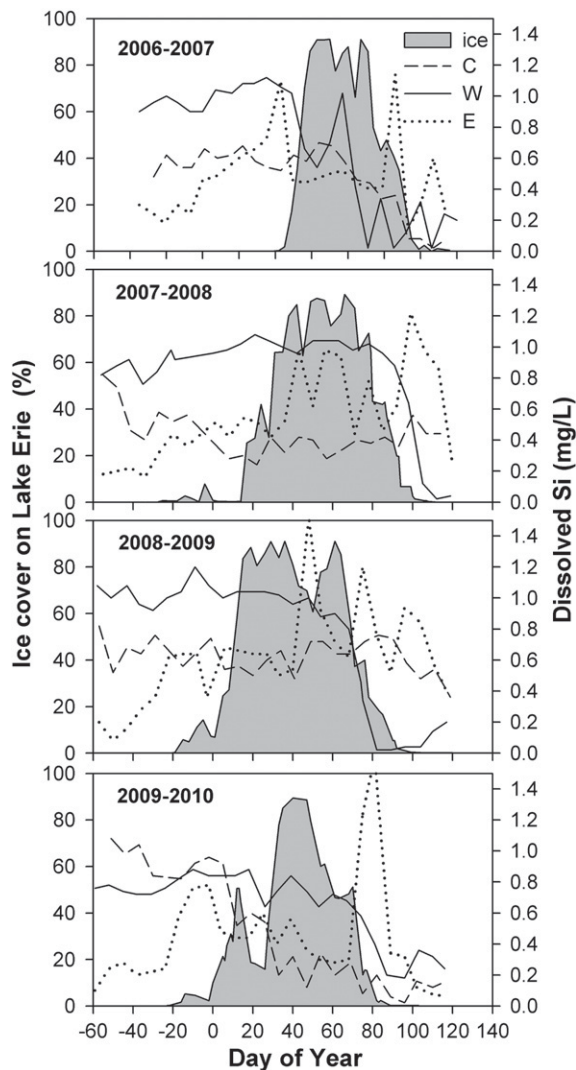


Fig. 2. Lake wide ice coverage (grey shading) and dissolved reactive silicate in water collected from nearshore regions along the north shore of Lake Erie from early November to late April during the study period. Ice cover refers to the percentage of the lake surface that is at least 10% covered in ice. Symbols represent the central (C), west (W) and east (E) basins of the lake.



Fig. 3. Extensive colouration of surface water directly below ice cover in the middle of the central basin of Lake Erie (February 2008) made visible by the passage of the ship. Coloration is due to the presence of high concentrations of the filamentous centric diatom *Aulacoseira islandica* and silt. Note the distinct boundary between colored water (brown) under translucent white ice and clear water (blue) under black (transparent) ice (lower right of image).

wake. Based on the 2008 surface ice survey, caches were present (mean $\pm$ SD) throughout the western basin (8.8 ± 4.2 caches per km of ship track) and central basin (6.2 ± 2.9 per km), and ranged from approximately 10–2000 m² in size (assuming circular shapes). One cache in the central basin (Fig. 3) extended for a distance of 7.4 km – the entire distance of white ice found between two leads. Caches were observed in areas of the lake that are greater than 20 m deep, thus it is unlikely that the silt and biomass observed were due to resuspension by ship propeller wash. Upon melting, sediment (silt) settled from some slush collected from caches; silt was noticed in the bottom of carboys (0.5 m deep) that were left at laboratory temperatures (5–10 °C) for two to four hours.

Lake Physics

The temperature depth profiles of the master stations (or stations adjacent to them; Fig. 1) in each of the lake basins commonly revealed weak inverse stratification with water temperatures less than 4 °C during February of 2008 and 2009 (Fig. 4A–C and G–I). Water column profiles measured in January, 2009 (Fig. 4D–F) revealed the lake to have a gradient with warmer water (1 to 1.4 °C) in the deeper east basin and cooler water in the west (0.14–0.16 °C). Thermal structure in the water and in situ Chl-*a* showed no concomitant patterns and oxygen levels were supersaturated (solubility of O₂ at 0 °C = 14.62 mg/L), except at Sta. 452 in January, 2009 ($\geq 95\%$ saturation). The slight clinograde DO profiles in the east basin (Fig. 4C, F, I) on all sampling dates appear to be related to the effect of inverse stratification and the lesser solubility of O₂ in warmer temperatures with increasing depth.

The data obtained for secchi depths and light extinction coefficients of PAR (Table 1) reflect highly turbid waters during most sampling periods and suggest a very shallow photic zone depth. A linear least squares regression of the natural logarithm of wavelength-specific extinction coefficients obtained at each station (Table 2) versus wavelength from 305 nm to 399 nm provides an estimate of *S*, the spectral slope parameter. *S* ($\pm$ standard error) was 0.0072 ± 0.00073 at Sta. 84, 0.0100 ± 0.00095 at Sta. 23, and 0.0117 ± 0.00114 at Sta. 1026. Light attenuation was strong, even in visible wavebands and PAR. The ambient DOC was quite chromophoric and had low spectral slopes. DOC concentrations measured at 11 stations in 2009 in 2010 indicate relatively invariant concentrations (20.0 ± 3.2 mg/L, mean $\pm$ SD, $n = 15$) compared to CDOM

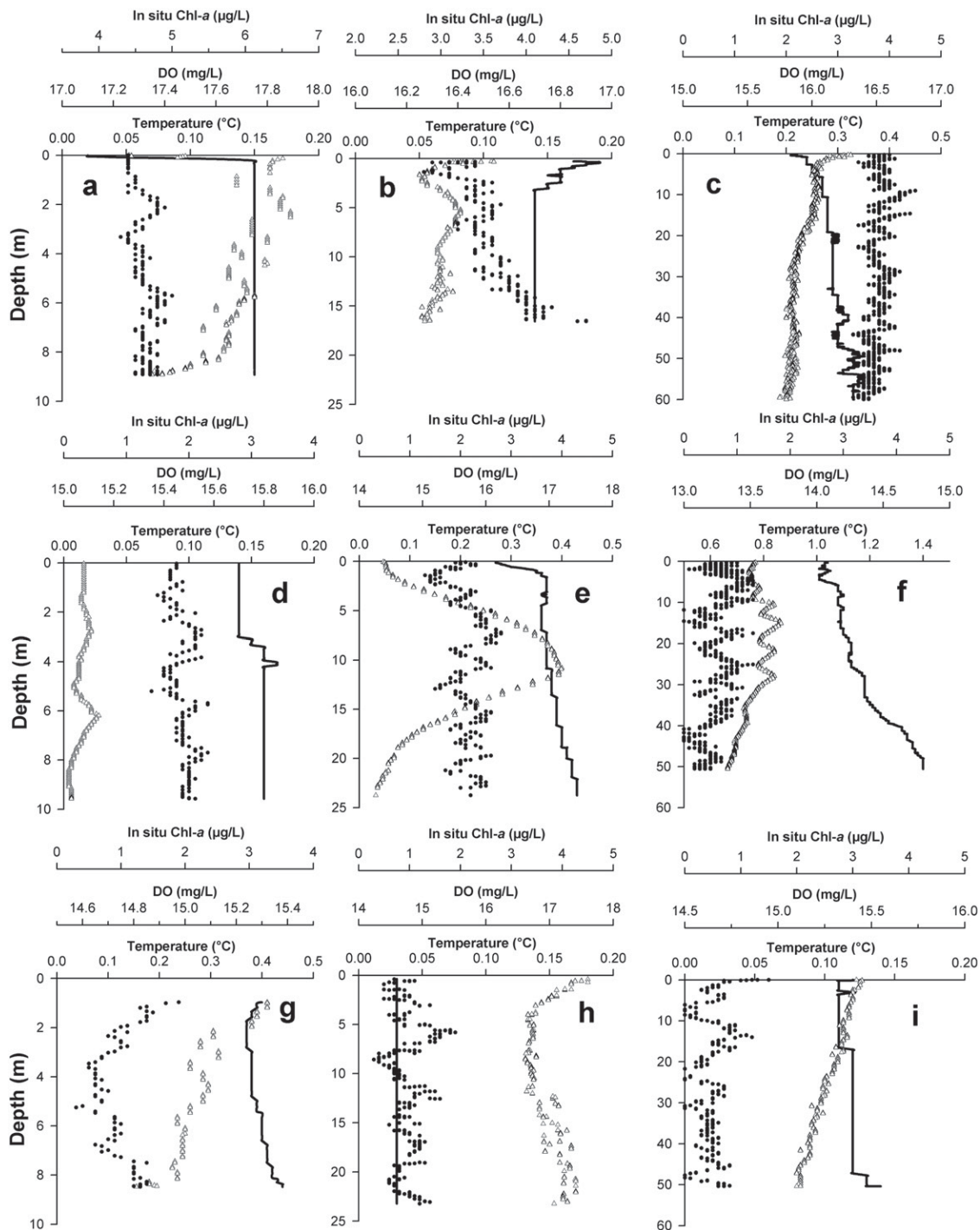


Fig. 4. Water quality profiles at master stations representing each of the three basins of Lake Erie (west basin, Sta. 357 [A, D, G]; central basin, Sta. 84 [B, E, H]; east basin Sta. 452 [C, F, I]) during February 12–16, 2008 (A–C), January 13–15, 2009 (D–F), and February 17–19, 2009 (G–I). Solid line = °C, triangles = dissolved oxygen (DO), solid circles = chlorophyll-*a*. Note the differing depth scale among stations; abscissa scales differ for clarity of profile traces.

that varied up to two-fold (Table 3). CDOM measured in all basins was greater in winter (Table 3) than that measured across the lake in summer (e.g. median CDOM at 1 m depth = 0.44 mg/L, $n = 3,495$ measurements across Lake Erie during July 2005, M.R. Twiss, unpublished data). The relative magnitude of attenuation coefficients for UV versus PAR and blue-green light suggested a significant influence from inorganic particles and a moderate concentration of CDOM at most stations (Sta. 340 may have had a greater CDOM concentration). Some caution of interpretation is merited because deployment conditions were much less than ideal except possibly at Sta. 1026.

Plankton Assemblage

Components of the phytoplankton assemblage varied considerably among the lake stations sampled, but overall, the community was dominated by phototrophic microplankton, primarily comprising filamentous diatoms (Table 4). In 2008, diatoms comprised 75–94% of the phytoplankton carbon. In terms of biomass, *Aulacoseira islandica* (Fig. 5) accounted for > 60% of microplankton assemblage at all locations in 2007 (Fig. 6) and 2008 (Fig. 6, Table 4). In caches observed at Sta. 84, the abundance of *A. islandica* was nearly an order of

Table 1

Secchi depth and light extinction coefficients (in parentheses) of photosynthetically active radiation (1/m) measured at study stations on Lake Erie, during winter from 2007 to 2010. NA = no data available.

Station	Date				
	Feb. 21–22, 2007	Feb. 12–16, 2008	Jan. 13–15, 2009	Feb. 17–19, 2009	Feb. 17–18, 2010
357	3	1	2	7	NA
340	3.2	1 (1.32)	NA	NA	NA
341	NA	NA	NA	NA	2.5 (0.95)
1026	NA	1.5 (0.78)	NA	3.5	NA
1290	NA	NA	NA	NA	1.2 (0.91)
84 (880)	NA	1 (1.60)	1.2	1.2	1.2 (1.23)
1053	NA	NA	NA	1.5	NA
452	NA	NA	1.3	4.5	3 (0.45)
23	4.9	1 (1.32)	NA	NA	NA

Table 2

Vertical light attenuation coefficient $\pm$ standard error (n) by wavelength for stations in Lake Erie during February 12–16, 2008. NA = no data available.

λ , nm	Sta. 340	Sta. 1026	Sta. 84	Sta. 23
665	NA	1.060 ± 0.007 (43)	1.730 ± 0.011 (48)	1.382 ± 0.009 (33)
554	NA	0.627 ± 0.005 (64)	1.443 ± 0.012 (51)	1.048 ± 0.008 (33)
510	NA	0.707 ± 0.006 (52)	1.614 ± 0.013 (51)	1.198 ± 0.007 (33)
491	NA	0.730 ± 0.004 (52)	1.848 ± 0.021 (31)	1.269 ± 0.007 (33)
443	NA	0.961 ± 0.004 (43)	2.065 ± 0.018 (38)	1.624 ± 0.010 (33)
399	NA	1.268 ± 0.004 (42)	2.864 ± 0.027 (31)	2.253 ± 0.021 (29)
379	NA	1.436 ± 0.005 (31)	3.202 ± 0.025 (30)	2.583 ± 0.028 (29)
340	5.683	2.231 ± 0.005 (204)	4.031 ± 0.015 (141)	3.615 ± 0.023 (130)
320	4.742 ± 0.031 (91)	2.794 ± 0.011 (131)	4.533 ± 0.030 (130)	6.675
305	5.874 ± 0.062 (94)	3.936 ± 0.172 (35)	6.004 ± 0.190 (89)	9.563
PAR	1.604 ± 0.006 (145)	0.780 ± 0.002 (207)	1.321 ± 0.006 (115)	1.318

Table 3

Chromophoric dissolved organic matter (CDOM) at various water depths sampled in Lake Erie in February 2007 and 2010. Values of CDOM (mg/L) are Suwannee River fulvic acid equivalents.

Station	Feb. 21–22, 2007	Feb. 17–18, 2010
357	1 m = 2.01, 5 m = 2.06, 8 m = 2.67	ND
340	1 m = 1.13, 10 m = 1.10	ND
341	ND	1 m = 2.72
84 (880)	1 m = 1.31, 20 m = 1.15	1 m = 1.05
1290	ND	1 m = 3.50
1053	ND	1 m = 1.15, 18 m = 1.09
23 (452)	1 m = 1.31, 30 m = 1.26	1 m = 2.54, 52 m = 2.37

magnitude greater than measured in surface waters at other sites in 2007 and 2008 (Fig. 6). In 2007, *A. islandica* accounted for >80% of the microplankton component, whereas other taxa (*Stephanodiscus*

binderanus (Kützing) Krieger, *S. niagarae* Cleve-Euler, *Fragilaria* spp., *Asterionella formosa*, and *Cyclotella* spp.) were subdominant, yet routinely observed. Other phytoplankton groups made up considerable amounts of biomass at times: e.g. *Scenedesmus* spp. represented 3720 $\mu\text{m}^3/\text{mL}$ at a depth of 10 m in Sta. 340, members of the Pyrrophyta composed 49,511 and 1428 $\mu\text{m}^3/\text{mL}$ at stations 1026 and 84, respectively, in 2008, and cyanobacteria accounted for 25,917 $\mu\text{m}^3/\text{mL}$ (59% of the particulate carbon) at Sta. 357 in 2008 whereas diatoms composed only 11% at this station.

Chl-*a* concentrations in caches characterized near Sta. 84 were 20.5 $\mu\text{g/L}$ in 2007 and 45.2 $\mu\text{g/L}$ in 2008. Ice collected from above caches that contained visible material had greater phytoplankton biomass than in the underlying lake water: e.g., ice collected near Sta. 949 in 2008 had 73.8 $\mu\text{g Chl-}a/\text{L}$ whereas the underlying surface water was 7.5 $\mu\text{g Chl-}a/\text{L}$, an additional cache sampled approximately 2 km from Sta. 84 in 2008 had ice with 23.7 $\mu\text{g Chl-}a/\text{L}$ whereas the underlying water was 11.8 $\mu\text{g Chl-}a/\text{L}$.

Phototrophic picoplankton (Ppico) abundance ranged 1-order of magnitude in the open water (10^6 to 10^7 cells/L), and nearly 2-orders of magnitude (10^6 to 10^8 cell/L) in the ice samples (Table 4). The assemblage was composed of cyanobacteria, namely single-celled cocci (e.g., *Synechococcus*), or small colonies of coccoid cells. One difference was the occurrence of picoeukaryotes (e.g., *Chlorella*) that were present in these samples, although typically at densities less than 10^5 cells/L. Phototrophic nanoplankton (Pnano) varied an order of magnitude among samples, and were present in lesser numbers compared with Ppico. This assemblage was dominated by cryptophytes (*Rhodomonas minuta*, Skuja, *R. lens* Pascher et Ruttner) and the ratio of phototrophic versus heterotrophic (non-bacterial) cell abundance in the water column ranged from 6.8 to >40.

Phytoplankton biomass, inferred from total Chl-*a* concentrations was consistently greatest in the transition area from west to central

Table 4

Abundance of phototrophic plankton size classes determined from samples collected at 5 lake stations and from collections of floating ice. Organisms were enumeration from water (1 m depth) and ice samples (water from thawed floating ice) collected in February 2008. Ice samples are from separate collections at Sta. 84.

Sample	Picoplankton (0.2–2 μm ; cells/L)	Nanoplankton (2–20 μm ; cells/L)	Phototrophic microplankton (20–200 μm ; cells/L)
357-water	$2.56 \pm 0.17 \times 10^7$	$2.63 \pm 0.17 \times 10^5$	$6.06 \pm 0.79 \times 10^4$
340-water	$1.50 \pm 0.11 \times 10^6$	$1.57 \pm 0.11 \times 10^5$	$8.37 \pm 0.53 \times 10^5$
1026-water	$3.69 \pm 0.20 \times 10^6$	$2.47 \pm 0.17 \times 10^5$	$8.16 \pm 0.52 \times 10^5$
84-water	$1.02 \pm 0.06 \times 10^6$	$5.00 \pm 0.34 \times 10^4$	$2.71 \pm 0.18 \times 10^5$
23-water	$1.44 \pm 0.10 \times 10^6$	$1.03 \pm 0.07 \times 10^5$	$1.63 \pm 0.13 \times 10^5$
84-ice, no. 1	$1.07 \pm 0.07 \times 10^8$	$3.60 \pm 0.25 \times 10^4$	$1.31 \pm 0.12 \times 10^6$
84-ice, no. 2	$4.83 \pm 0.34 \times 10^7$	$3.12 \pm 0.21 \times 10^5$	$1.63 \pm 0.14 \times 10^6$
84-ice, no. 3	$1.87 \pm 0.12 \times 10^6$	$2.40 \pm 0.17 \times 10^4$	$3.61 \pm 0.23 \times 10^5$
84-ice, no. 4	$2.17 \pm 0.15 \times 10^6$	$6.10 \pm 0.43 \times 10^4$	$4.38 \pm 0.24 \times 10^5$

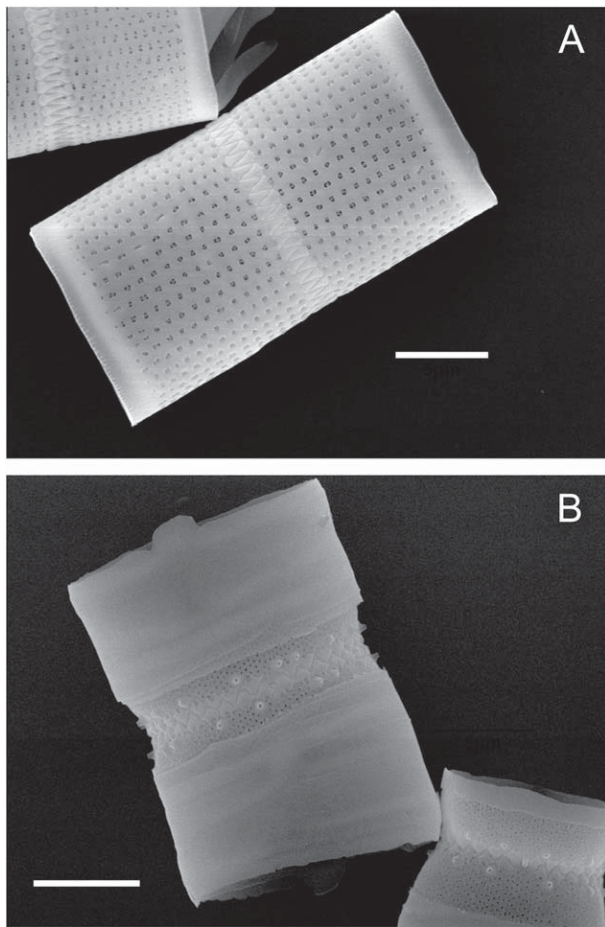


Fig. 5. Scanning electron microscopy (SEM) images of diatom frustules present in ice-covered Lake Erie. A. The microplankton was dominated by the filamentous diatom *Aulacoseira islandica*. B Other notable genera included members of *Stephanodiscus* spp., particularly *S. binderanus* (shown), which accounted for about 20% of the microplankton at western and central basin stations. Scale bars = 5 μ m.

basin at approximately 82.4° west longitude (Fig. 7A), and tended to decrease with distance east. Consistent with the eastward decrease in Chl-*a* was a decrease in dissolved nutrient levels in nitrate + nitrite (Fig. 7B) as well as silicate (Fig. 7C). Unlike dissolved nutrient concentrations, total phosphorus concentrations ($18.5 \pm 5.7 \mu\text{g/L}$, mean $\pm$ SD) did not show an eastward decreasing trend on most surveys (Fig. 7D). No APA was detected in lakewater (data not shown) suggesting sufficient bioavailability of phosphorus to plankton. POC concentrations in surface water (Fig. 8A) reflected the longitudinal trend observed for total Chl-*a* (Fig. 7A). Seston stoichiometric ratios revealed N:P molar ratios that were generally <16 (Fig. 8) whereas median C:N ratios (9) were uniformly, though only marginally, greater than the Redfield ratio of 6.6.

Members of the heterotrophic plankton were present in appreciable numbers in the lake, but exhibited considerable variation among stations (Table 5). Bacterial abundances in the water column and in two ice samples ranged from 10^9 to 10^{10} cells/L. Bacteria were observed associated with extracellular polymeric substances on the surfaces of filaments of *Stephanodiscus* and *Aulacoseira* (data not shown). The abundance of heterotrophic nanoflagellates varied by one order of magnitude among open lake sites (10^5 to 10^6 cells/L); their abundances were similar between open water and ice assemblages. Crustacean zooplankton densities were low and dominated by copepods,

which comprised $92 \pm 4\%$ of the crustacean zooplankton population; cladocera made up the remainder.

Physiological Status of Phytoplankton

Measures of dark-adapted F_v/F_m indicated that the maximum quantum efficiency of PSII was high (median = 0.55 of five stations) in surface water samples collected from western and central basin stations in February 2008. Likewise, samples collected directly from caches in the vicinity of Sta. 84 ($F_v/F_m = 0.60$), from a vertical net tow at this same station ($F_v/F_m = 0.56$) and from ice-melt samples from central basin Sta. 949 ($F_v/F_m = 0.54$) demonstrated high F_v/F_m . Only the sample from eastern basin Sta. 23 ($F_v/F_m = 0.40$) and an ice-melt sample from near Sta. 84 ($F_v/F_m = 0.33$) in the central basin had relatively low F_v/F_m (≤ 0.4) indicating sub-optimal physiological condition and suggesting that these samples were stressed or contained senescent algal cells. Measurements in 2009 also revealed $F_v/F_m > 0.5$ in phytoplankton samples at these stations and further showed that algae sampled *in situ* in unmelted ice using a PhytoPAM could have similarly high F_v/F_m .

Silicate

Decreases in silicate concentrations from west to east was evident in offshore areas of the lake during all five cruises (Fig. 7C) and in nearshore areas in the west basin as the winter progressed during all study years and in the central basin in 2007 and 2010 (Fig. 2); no Si drawdown was observed in the water intake data from the east basin station. One caveat to the nearshore data interpretation is that the nearshore areas can be affected by ice cover; during winter the west basin intake at Union, ON is under ice cover more so than the central and east basin intakes thus, it is less prone to wind-driven mixing that could increase water column concentrations of silicate. If reduction of silicate concentrations in the nearshore region is related to diatom biomass production in offshore regions then the cruise dates (day of year 13–50) coincided with the beginning of the period of potential silicate depletion.

Discussion

Lake Erie in Winter

The present observations offer us the opportunity to examine the winter limnology in Lake Erie over a time span of nearly seven decades. Chandler's observations of the west basin in the late 1930s are consistent with what we have observed here, namely increased water clarity under extensive ice cover in the west basin, and isothermal conditions with the water column fractions of a degree above freezing (Chandler, 1940, 1942a, b). Stewart (1973) also reported similar temperature-depth profiles as we have observed, and reasoned that extensive inverse stratification did not occur in the lake due to opening of leads in the ice pack that allows for wind driven mixing.

Wind driven mixing likely has an important influence on development of a winter diatom bloom in Lake Erie. High turbidity both under ice and in open water reaches was consistently observed in the central basin of the lake, together with highly chromophoric DOC that had spectral absorption slopes more typical of values we have observed in rivers than in the Great Lakes. This is presumably due to wind-driven sediment resuspension into an isothermal water column and the retention of fine grained material (CDOM, clay) in the water column even after the establishment of ice cover. Nicholls (1998) demonstrated that total phosphorus concentration in winter was negatively correlated with the maximal percentage of Great Lakes surface area covered by ice, consistent with a role for wind-aided suspension of nutrients including phosphorus (and sediment) in ice-free water during winter.

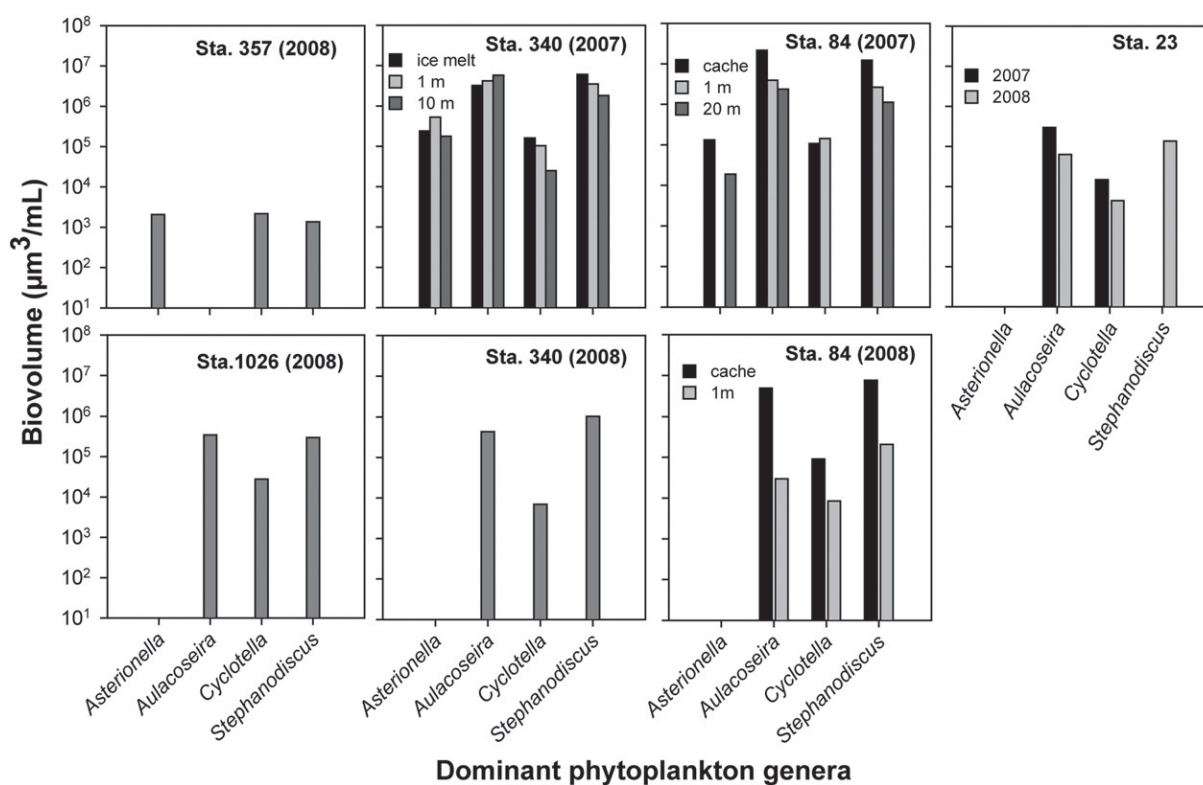


Fig. 6. Microplankton and nanoplankton biomass (cell volume) from samples collected across Lake Erie during cruises in February 2007 and 2008. All samples were collected at a depth of 1 m unless noted.

Nutrients are affected by biological activity in winter as well as physical processes such as wind-driven mixing. Silica concentrations in Lake Erie in winter reflect phytoplankton dynamics. Silica concentrations measured in spring in Lake Erie are much greater now than before colonization by dreissenid mussels, presumably due to diminished primary production demand for Si as a result of high grazing losses. Nevertheless, silicate concentrations are reduced even during the winter because of the appreciable diatom biomass, and presumed Si demand, shown here. Chandler (Chandler, 1940, 1942a,b) reported the dominance of diatoms in the plankton under ice. Likewise, later (40 years) investigations of Lake Erie in January and February reported a dominance of diatoms in the lake phytoplankton at this time (Burns et al., 1978), a finding which we confirm. A comparison of Chl-*a* concentrations from 1985 to present across all basins does not give convincing evidence of systematic change (Fig. 9A), but the more recent data are more abundant and reveal a much wider range of concentrations (see below). Silicate concentrations in the period of 2007–2010 were approximately 10-fold greater than in 1985 and earlier (Fig. 9B). By contrast, there was no apparent trend in total phosphorus concentrations in winter over the same 40 year period (Fig. 9C). Winter levels of N and P chemical forms recorded from nearshore sites sampled as part of the OME Great Lakes Water Intake Monitoring program (Nicholls et al., 2001; Ontario Ministry of Environment and Energy, 1997), and average post-*Dreissena* spring levels of nutrients reported by Makarewicz et al. (2000), are comparable to those shown here. The markedly greater levels of dissolved silica measured during the present study are consistent with the reported ecosystem engineering effects of *Dreissena* mussels in Lake Erie (Barbiero et al., 2006; Holland et al., 1995; Makarewicz et al., 2000). Alleviation of silicate demand outside of the winter period may now provide winter diatoms with a greater

initial (late autumn-early winter) concentration of silicate so their demands do not deplete silicate to the lesser concentrations observed in earlier decades.

Prevalence of Diatoms During Winter

The presence of an intense winter phytoplankton bloom of cold-tolerant diatoms was evident from the elevated Chl-*a* concentration in the lake at this time of year (particularly in some caches that contain over 100 μg Chl-*a*/L; Smith, 2011). Smaller non-diatom phytoplankton are also evident in surface waters in winter. Abundances of autotrophic picoplankton cyanobacteria and nanoflagellates at stations 84 and 23 tend to be more typical for this time of year based on previous sampling in Lakes Huron and Michigan (Fahnenstiel and Carrick, 1992; Carrick and Fahnenstiel, 1989, respectively). The occurrence of a similar morphotype of autotrophic picoplankton has been observed in the upper Great Lakes in the late winter and spring period (Fahnenstiel and Carrick, 1992; Fahnenstiel et al., 1986).

Aulacoseira islandica was the dominant taxon found in the micro-phytoplankton size-fraction during the surveys. In a related study (Feb. 2010) using Si-deposition specific dye, Saxton et al. (2012/in revision) noted that *Aulacoseira islandica* was the dominant diatom in winter samples (with about 90% of the filaments showing active Si deposition) but the phytoplankton community biomass in the summer of 2010 was dominated by *Fragilaria* spp. and *Asterionella* spp.. Whereas *A. islandica* has long been identified a component of the spring diatom assemblage in the Great Lakes, including Lake Erie (Barbiero and Tuchman, 2001; Makarewicz, 1993; Stoermer et al., 1981) and this species was detected in Lake Erie in November and December 1970 when it composed 10.5% and 8.5% of the phytoplankton biomass (Munawar and Munawar, 1976), it has emerged as a

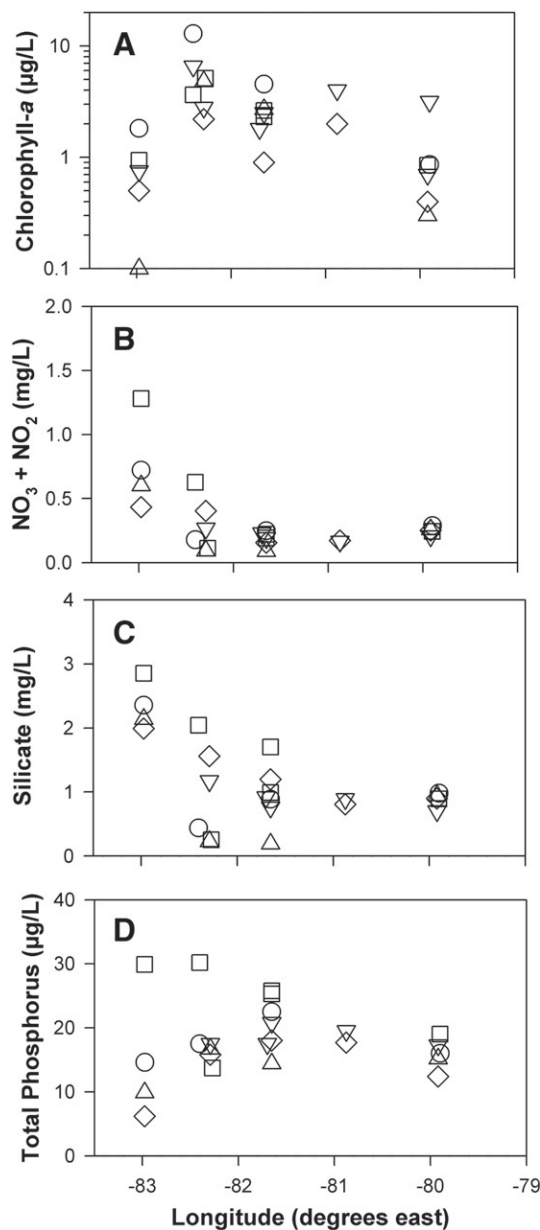


Fig. 7. Chlorophyll-*a* (>0.2 µm) and nutrient concentrations at a depth of 1 m at fixed stations in Lake Erie during periods of extensive ice cover over the entire lake. Symbols: ○ = Feb. 21–23, 2007; □ = Feb. 12–16, 2008; △ = Jan. 13–15, 2009; Δ = Feb. 15–17, 2009; ▽ = Feb. 16–18, 2010. Note the logarithmic scale for Panel A.

dominant spring species only since the early 1990s, a shift coinciding with the invasion of dreissenid mussels in Lake Erie (Barbiero et al., 2006). In the 1998 EPA-GLNPO spring survey, *A. islandica* accounted for 77% of all diatom biomass and 69% of all phytoplankton biomass in Lake Erie (Barbiero and Tuchman, 2001). The changes to Lake Erie following the dreissenid mussel invasion are widespread and well-documented and include shifts in nutrient cycling as well as changes in biomass and composition of the phytoplankton community (Barbiero et al., 2006; Makarewicz et al., 2000; Nicholls and Hopkins, 1993). Particularly relevant to the emergence of *A. islandica* as a dominant species during vernal mixis has been the dramatic increase in silica levels measured during spring with averages 10-fold greater than pre-*Dreissena* concentrations (Fig. 9B). Pennate diatoms such as *Asterionella formosa*, *Fragillaria crotonensis* and *Tabellaria flocculosa* (Roth) Kützing that dominated the pre-*Dreissena* spring assemblage possess relatively low calculated Si optima of <0.2 mg SiO₂/L (Barbiero et al., 2006). By contrast, *A. islandica* along with *Stephanodiscus parvus*

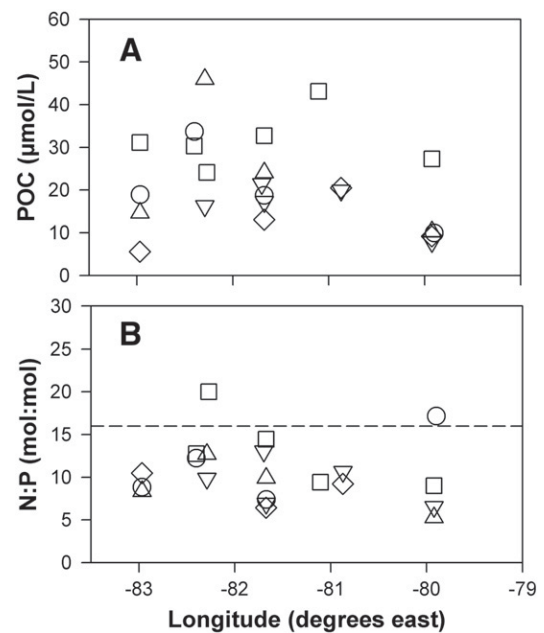


Fig. 8. Particulate organic carbon (POC) and the nitrogen:phosphorus ratio of seston collected at a depth of 1 m at fixed stations in Lake Erie during periods of extensive ice cover over the entire lake. Symbols: ○ = Feb. 21–23, 2007; □ = Feb. 12–16, 2008; △ = Jan. 13–15, 2009; Δ = Feb. 15–17, 2009; ▽ = Feb. 16–18, 2010. Dashed line in Panel B refers to the Redfield N:P ratio of 16.

Stoermer & Håkansson and *S. hantzschii* Grunow, centric diatoms reported to co-dominate the spring assemblage in the eastern basin of Lake Erie (Carrick, 2004), are characterized as high silica-requiring species with calculated Si optima exceeding 0.6 mg SiO₂/L (Barbiero et al., 2006).

Despite its characterization as a cold stenothermal species (Stoermer and Ladewski, 1976), it has not been determined whether *A. islandica* that emerges during winter and persists at least through spring in Lake Erie is an obligate psychrophile. That it is no longer present in the lake in August (Barbiero and Tuchman, 2001) is consistent with this characterization; however, verification of this trait awaits the isolation and successful culture of this taxon. Physiological resting cells of diatoms, including *A. islandica* have been identified from surficial sediments in the Great Lakes (Lashaway and Carrick, 2010; Sicko-Goad et al., 1989). Lashaway and Carrick (2010), reported the meroplanktonic assemblage of Lake Erie to be dominated by *A. islandica*, *S. niagarae* and *S. binderanus* and that most of the cells collected from surficial sediments were vegetative resting cells. Their potential to seed seasonal diatom blooms, including the bloom we describe for winter has been demonstrated via sediment

Table 5

Abundance of heterotrophic plankton by size classes as determined from samples collected at 5 lake stations and from collections of floating ice. Organisms were enumeration from water (1 m depth) and ice samples (water from thawed floating ice) collected in February 2008. Ice samples are from separate collections at Sta. 84. Crustacean zooplankton data are from vertical tows in 2007.

Station	Heterotrophic picoplankton (0.2–2 µm; cells/L)	Heterotrophic nanoplankton (2–20 µm; cells/L)	Crustacean zooplankton (> 153 µm; individuals/L)
357	$3.0 \pm 0.05 \times 10^{10}$	$5.52 \pm 0.36 \times 10^5$	8.23×10^{-2}
340	$4.1 \pm 0.11 \times 10^{10}$	$1.06 \pm 0.07 \times 10^6$	4.59×10^{-1}
1026	2.0×10^{10} (n=1)	$1.18 \pm 0.08 \times 10^6$	ND
84	$3.7 \pm 1.4 \times 10^{10}$	$7.73 \pm 0.54 \times 10^5$	3.53×10^{-1}
23	$1.3 \pm 0.04 \times 10^9$	$7.43 \pm 0.52 \times 10^5$	2.00×10^{-1}
Ice 1	$5.5 \pm 0.02 \times 10^{10}$	$2.39 \pm 0.17 \times 10^6$	ND
Ice 2	$2.4 \pm 0.16 \times 10^9$	$2.85 \pm 0.20 \times 10^6$	ND
Ice 3	ND	$4.44 \pm 0.30 \times 10^5$	ND
Ice 4	ND	$3.98 \pm 0.28 \times 10^5$	ND

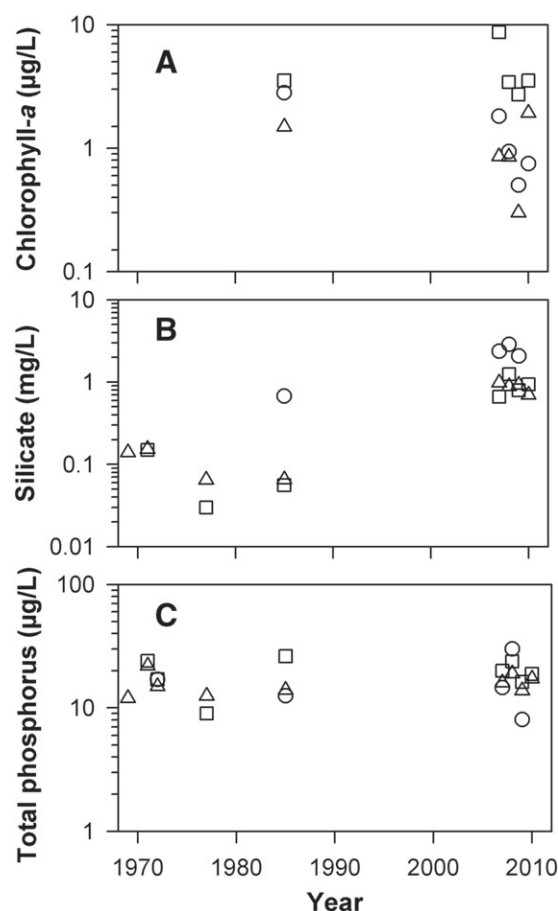


Fig. 9. A comparison of historical and present offshore surface water quality data among basins in Lake Erie in the months of January and February. Values are averages of all data collected per basin for a given year; data for 1969, 1971, and 1977 are from Burns et al. (1978); data from 1972 are from Stewart (1973); data for 1985 from are Rockwell et al. (1989); data for 2007–2010 are from this study. Symbols: circle (west basin), square (central basin), triangle (east basin).

resuspension assays, the success of which is affected by light, temperature and possibly nutrient conditions (Lashaway and Carrick, 2010; Sicko-Goad et al., 1989).

Despite supporting moderate to high Chl-*a* biomass, the lake was not characterized as nutrient deficient during the winter expeditions. Seston stoichiometric ratios were consistent with a P-sufficient assemblage with N:P ratios below Redfield stoichiometry. Light limitation can alter the N:P ratio for balanced growth considerably compared to the Redfield ratio (e.g. Finkel et al., 2006) but the lack of community alkaline phosphatase activity further supported the absence of P deficiency. Variable fluorescence (F_v/F_m) also indicated a high quantum efficiency of PSII in almost all samples whereas deficiency of P or N can reduce the quantum efficiency (e.g. Geider et al., 1993). Inasmuch as the phytoplankton assemblage showed indication of moderate N deficiency based on seston C:N ratios (Healey and Hendzel, 1980), mean C:N ratios parallel those reported recently for large lakes (C:N = 9.2, Sterner et al., 2008; and C:N = 7.1 to 8.6 in east basin of Lake Erie in summer, North et al., 2007).

Aside from their potential role in seeding seasonal diatom blooms, Carrick et al. (2005) identified meroplanktonic resting cells that accumulate in the benthos, which may also contribute to the formation of recurring summer hypoxia in Lake Erie's central basin. Carrick et al. (2005) demonstrated that a large fraction of the organic matter present in the benthos is composed of diatoms (e.g., *Aulacoseira islandica*, *Stephanodiscus binderanus*, *S. niagare*) that are present during early season blooms in the lake. More recently, Lashaway and Carrick (2010) demonstrated that the oxidation of algal matter sampled

from surface sediments at three locations in the central basin of Lake Erie can contribute from 11 to 33% of oxygen depletion recorded at these sites during summer. In our study we found that bacterial abundance in the lake was similar to densities observed during the summer (DeBruyn et al., 2004). However, recent observations by our group (unpublished data) suggest that rates of microbial carbon production in the winter are much less than those in the summer: similar observations have been made in other coldwater systems (Pomeroy and Deibel, 1986) and perhaps represent a major cause of the seasonal disconnects between primary and secondary carbon production in cold water systems (Kirchman et al., 2009). The direct contribution of winter diatom production to summer hypoxia in Lake Erie's central basin remains to be elucidated. Either way, loss of oxygen in the hypolimnion of the central basin of Lake Erie in the summer is well documented (Bertram, 1993; Charlton et al., 1993) and a function of the active microbial communities in this lake during summer months. When expansive, the area can exceed 10,000 km² (Hawley et al., 2006), comparable in surface area to the low oxygen 'dead zone' in the Gulf of Mexico.

Plankton Assemblages Associated with Ice

Whereas caches were visually distinct features in each of the five surveys conducted, they were evident as discrete accumulations of *Aulacoseira* biomass under the ice rather than as a widespread dispersed bloom. Mean winter Chl-*a* levels recorded from Lake Erie were comparable to those reported more recently during the spring isothermal period in the lake (Barbiero and Tuchman, 2001; Makarewicz et al., 2000). When specifically targeted, however, cache sites routinely supported Chl-*a* levels > 20 µg/L consistent with the brown color of the water. Caches with Chl-*a* as high as 154 µg/L in February, 2010 and 97 µg/L in February, 2011 were documented with the central basin containing the highest and most variable concentration of Chl-*a* (27 ± 45 µg/L in 2010, 21 ± 31 µg/L in 2011; mean $\pm$ SD) of all basins as determined by hourly sampling of surface waters while traversing the entire lake (Smith, 2011). At all sites in the present study, the microphytoplankton size class (> 20 µm) represented 60% or more of Chl-*a* biomass; this is consistent with the findings of the 1998 U.S. EPA's biological surveillance program report of diatom contribution to > 80% of the phytoplankton biomass in offshore waters of Lake Erie during early April (Barbiero and Tuchman, 2001) as well as a recent monitoring surveys facilitated by partnership with the U.S. Coast Guard (McKay et al., 2011). In lake wide sampling of surface waters in February, Smith (2011) documented that 14% of surface water samples in 2010 and 11% in 2011 exceeded 20 µg Chl-*a*/L; the 20 µg Chl-*a*/L concentration is considered the threshold for assessing the presence of cache based on these values > 20 µg Chl-*a*/L being outliers, where the upper outlier concentration was established as the upper quartile plus 1.5 times the interquartile range of the data. Thus, it appears that caches are a widespread phenomenon existing on Lake Erie during winter.

The high density of picoplankton in the ice was unexpected. These densities are considerable and rival those in other productive areas of the lake during the summer. The abundances observed in winter were comparable to that seen in Lake Erie in the offshore region of the lake during summer months (cf. Carrick, 2004; Wilhelm et al., 2006). Although cyanobacteria are known to proliferate in Lake Erie in summer, the presence of cyanobacteria at concentrations similar to summer concentrations is understandable given their cryotolerant but not cryophilic (psychrophilic) nature (Vincent, 2007).

How caches form, persist, and disperse is not known. However, based on the observations we have made over four years on Lake Erie we can speculate on their formation, distribution, fate, and biogeochemical importance to the lake. Silt was noticed in slush thawed from caches in the middle of the central basin of Lake Erie (approx. 20 m depth). Based on the settling velocity of fine (10 µm) to coarse (60 µm) silt of 8×10^{-5} m/s to 2.8×10^{-3} m/s (Hemond and

Fechner-Levy, 2000) it follows that silt would have 2 hours (coarse silt) to 70 hours (fine silt) to be retained in the top 1 m of the lake after a resuspension event. Wind-driven resuspension during isothermal conditions could bring silt to the surface and there is evidence that vertical circulation features such as Langmuir cells (Dethleff and Kempema, 2007) and release of positively buoyant anchor ice (Kempema et al., 2001) can lead to sediment entrainment in ice. Large lakes with discontinuous ice cover are also likely to experience frazil ice formation (Leppäranta and Wang, 2008), wherein particles (e.g. silt, phytoplankton [Table 4]) serve as nucleation sites for ice formation in super-cooled water in open leads on the lake surface that frequently form in large lakes. Particles trapped in positively buoyant ice could conceivably scavenge and trap larger particles such as microplankton (e.g. *Aulacoseira*), effectively explaining how silt could be found in surface ice at depths >20 m and how viable algae are found within ice. *Aulacoseira* and other plankton would proliferate directly since they are subject to a more favorable light environment and abundant nutrients, compared to lower in the water column below the ice. The extensive (>7 km wide) cache seen in the central basin in 2008 could have an origin in the storm that passed through the lake in the week preceding the 2008 investigation. A permanent NOAA-GLERL Real-Time Coastal Observation Network (ReCON; Ruberg et al., 2007) Station located at Toledo Channel Marker # 2 in the lake's western basin recorded maximum wind speeds of 50 knots from the north accompanied by a drop in air temperature of >10 °C to −20 °C just 2 days prior to the survey. The cache observed in the central basin (Fig. 3) was composed of a large amount of slush, consistent with the consolidation of frazzle ice. The presence of dense accumulations of *Aulacoseira islandica* (>150 µg Chl-*a*/L) in shallow fluvial Lake St. Pierre (St. Lawrence River) in mid-winter (Frenette et al., 2008) is also consistent with wind-driven entrainment of phytoplankton in ice. Note that attempts to estimate winter phytoplankton blooms in Lake Erie in winter by satellite (e.g. Kerfoot et al., 2010) must be tempered by the fact that it is below ice (and snow) cover however, the Envisat Medium Resolution Imaging Spectrometer with Maximum Chlorophyll Index signal is capable of detecting surface accumulations of algae within ice on Lake Erie (C.E. Binding, Environment Canada, pers. comm.).

Ice containing phytoplankton and silt will be more prone to structural decay due to disruption of the ice lattice by particles and the thermal corrosion due to the reduced albedo of the ice and the concomitant absorption of solar heat. Thus, the caches and their phytoplankton community dominated by *Aulacoseira* sp. may play a role in the structural stability of the ice sheet in Lake Erie. Climate change will have an impact on the extent and degree of ice cover in Lake Erie. Ice cover reduces sediment resuspension and distribution of internal nutrient loading (Nicholls, 1998) and may reduce primary productivity by reducing light flux to the water column. Thus, changes in ice cover will likely have an impact on the thermal regime in Lake Erie, the amount of organic carbon production in the lake and accordingly, dissolved oxygen levels. Predicting those future impacts requires winter limnology to be pursued.

A common notion is the presence of a spring bloom in Lake Erie. However, lake wide winter concentrations of Chl-*a* are significantly ($P<0.05$) greater than lake wide spring Chl-*a* concentrations based on lake wide transects collecting surface samples at hourly intervals conducted as part of this larger study: winter 2011 = 18 ± 36 , winter 2010 = 13 ± 23 , spring 2010 = 3 ± 1.6 , and summer 2010 = 8 ± 8 ; values are µg Chl-*a*/L (mean $\pm$ SD; Smith, 2011). These data provide clear evidence that very high Chl-*a* concentrations and high variability of Chl-*a* concentrations exists in Lake Erie surface waters in winter and that the winter bloom of phytoplankton in this lake is far greater than the subsequent spring time proliferation of phytoplankton.

Since considerable plankton biomass accumulates in the lake in winter, with the potential for drawdown of silicate, one can conclude that a larger spring diatom bloom is not possible due to reduced

nutrient availability. Understanding the ecological functioning of Lake Erie may require incorporating the winter as an important period of primary production, just as a re-examination of the North Atlantic phytoplankton bloom by Behrenfeld (2010) has revealed that blooms begin in winter not in the spring when the water column provides for maximum levels of mixing. To consider this model for winter bloom development in Lake Erie, the initiation of the winter bloom likely occurs during the isothermal conditions that persist from the breakdown of thermal stratification in early winter prior to the onset of ice cover. Seeding of the ice cover with phytoplankton, and extensive accumulations of phytoplankton in and below the ice (caches), through frazil ice accumulation and possibly other mechanisms such as anchor ice serves to support continued phytoplankton growth at the lake surface during ice cover, which is normally extensive on this lake. The accumulated phytoplankton biomass and high turbidity observed during the cruises described here may be considered the remnants of the winter bloom. Such high turbidity, along with the low winter insolation, would be expected to limit photosynthesis strongly, yet phytoplankton (and specifically diatoms) were clearly able to achieve moderate to high extracted Chl-*a* concentrations and not only near the surface, as evidenced by the rather uniform in situ Chl-*a* fluorescence profiles observed at most stations, an apparent paradoxical situation. This implies highly efficient light utilization by the phytoplankton, very low loss rates, or both, but reports of winter primary production, phytoplankton grazing, sedimentation or other component processes are scarce or absent for Lake Erie or comparable large lakes. Further communications will describe in greater detail processes amongst the microbial community structure and rates of metabolism in Lake Erie in winter.

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